

JYVASKYLA, Finland

WMO: 029350

Lat: 62.398N

Lon: 25.671E

Elev: 140

StdP: 99.66

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.4	-24.0	-30.1	0.2	-27.3	-26.4	0.3	-23.9	9.9	-3.3	8.5	-2.6	1.2	320	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	26.7	17.7	24.8	16.9	23.0	16.0	19.4	24.1	18.2	22.5	17.2	21.3	3.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	13.0	21.1	16.7	12.1	20.0	15.5	11.2	18.8	55.7	24.0	52.1	22.5	48.7	21.1	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.9	6.1	-30.2	28.9	4.2	2.0	-33.2	30.3	-35.7	31.5	-38.1	32.6	-41.1	34.0		
(5)			-30.3	20.6	4.2	1.6	-33.4	21.8	-35.8	22.8	-38.1	23.7	-41.2	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-8.1	-8.1	-4.1	2.4	8.5	13.4	16.5	14.6	9.4	3.6	-0.9	-5.0		
	DBStd	9.85	7.26	7.44	5.22	3.85	4.36	3.56	3.19	3.13	3.43	4.10	4.69	6.78		
	HDD10.0	2871	562	507	437	229	81	8	1	3	53	201	326	464		
	HDD18.3	5416	821	741	695	478	304	153	77	123	269	457	576	723		
	CDD10.0	527	0	0	0	1	35	111	201	144	33	2	0	0		
	CDD18.3	31	0	0	0	0	1	5	19	6	0	0	0	0		
	CDH23.3	329	0	0	0	0	20	64	186	59	0	0	0	0		
	CDH26.7	48	0	0	0	0	2	7	33	6	0	0	0	0		
(14) Wind	WSAvg	2.7	2.7	2.7	2.9	2.7	2.9	2.8	2.5	2.3	2.5	2.7	2.8	2.8		
(15) Precipitation	PrecAvg	631	43	31	33	35	42	61	79	81	60	60	55	49		
	PrecMax	846	82	85	70	79	114	157	150	182	128	129	114	94		
	PrecMin	397	12	3	3	2	5	8	4	16	12	14	4	13		
	PrecStd	94	18	18	14	19	23	29	37	40	26	28	24	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	4.8	9.6	18.6	25.9	27.8	29.4	27.2	20.9	13.7	9.5	6.7		
		MCWB	3.1	2.8	4.7	10.2	15.1	17.7	19.1	18.2	15.5	11.2	8.7	6.0		
	2%	DB	2.2	2.9	6.2	14.7	22.8	25.2	27.5	24.9	18.8	11.7	7.7	4.9		
		MCWB	1.7	2.1	2.6	8.4	13.5	16.2	18.7	17.4	14.3	10.6	7.1	4.4		
	5%	DB	1.3	1.6	4.2	11.9	19.9	23.0	25.3	23.2	16.8	10.8	6.2	3.3		
		MCWB	0.9	1.1	1.6	6.6	11.8	15.2	17.7	16.5	13.4	9.8	5.7	2.8		
	10%	DB	0.4	0.6	2.7	9.4	17.2	20.9	23.4	21.2	15.2	9.6	5.0	2.2		
		MCWB	0.2	0.2	0.9	5.3	10.5	14.2	17.0	15.9	12.6	8.7	4.5	1.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.2	5.2	11.8	16.6	19.4	21.4	20.2	16.2	12.1	8.8	6.1		
		MCDB	3.4	4.0	8.3	16.0	22.5	24.7	26.4	25.2	19.1	13.1	9.4	6.4		
	2%	WB	1.8	2.1	3.3	9.0	14.7	17.7	20.0	18.6	15.0	10.9	7.0	4.3		
		MCDB	2.1	2.8	5.5	12.9	19.9	22.3	24.8	22.7	17.4	11.5	7.4	4.7		
	5%	WB	1.0	1.1	2.2	7.2	13.0	16.4	18.9	17.5	14.1	9.9	5.7	2.9		
		MCDB	1.2	1.5	3.8	11.2	18.7	20.9	23.3	21.3	16.3	10.6	6.2	3.2		
	10%	WB	0.2	0.2	1.1	5.5	11.3	15.0	17.8	16.5	13.1	8.7	4.6	1.8		
		MCDB	0.4	0.5	2.2	8.9	16.1	19.7	21.9	20.1	14.8	9.3	4.9	2.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	7.0	8.8	9.4	11.0	10.1	9.9	10.0	8.5	5.4	4.1	5.0		
		MCDBR	4.4	5.8	10.0	14.3	15.8	14.0	13.1	13.1	11.2	6.5	4.9	3.9		
	5% WB	MCWBR	4.3	5.3	7.2	8.5	8.0	6.3	5.5	6.4	7.3	5.3	4.6	3.9		
		MCWBR	4.1	5.5	7.8	12.8	13.4	11.3	11.1	11.0	9.6	5.7	4.6	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.234	0.285	0.312	0.353	0.350	0.347	0.371	0.356	0.332	0.312	0.254	0.356		
		taud	2.042	2.145	2.212	2.336	2.447	2.478	2.427	2.460	2.517	2.415	2.140	2.636		
	Ebn at Noon		426	656	788	833	870	877	842	825	780	643	399	266		
		Edh at Noon	34	67	91	100	98	97	99	88	69	53	31	20		
(44) All-Sky Solar Radiation	RadAvg		0.19	0.75	2.24	3.64	4.84	5.21	4.97	3.84	2.20	0.89	0.23	0.10		
	RadStd		0.03	0.18	0.26	0.43	0.54	0.40	0.51	0.44	0.27	0.17	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.58	N/A	N/A	N/A	N/A	N/A	-192	-211	N/A	N/A
(47) Regional (45 neighbors)		+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page